Charge

1. Conservation – isolated system total electric charge
2. Quantization

   smallest unit of charge e → elementary
   ↳ proton or electron

   q ← variable

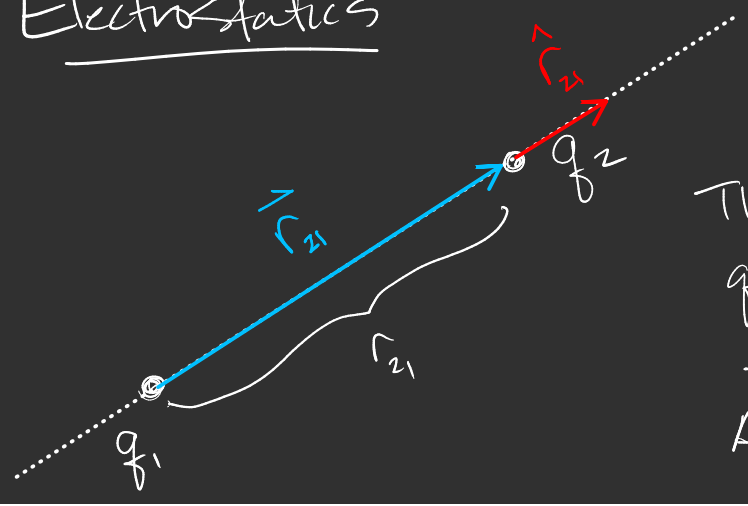
SI units → Coulomb $\quad 1e \equiv 1.602 \cdot 10^{-19}\,C$

$$\downarrow$$

$$1C = \frac{1}{1.602 \cdot 10^{-19}}\,e = 6.242 \cdot 10^{18}\,e$$

Electrostatics

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} \qquad r^2 = \vec{r} \cdot \vec{r}$$



The force between q_1 + q_2 is directly proportional to the product of q_1 + q_2 AND inversely proportional to distance squared btt them

Coulomb's Law

$$\vec{F}_{21} = \frac{k\, q_1 q_2}{r_{21}^2}\,\hat{r}_{21}$$

Coulomb constant

electric constant

$$k = 8.988 \cdot 10^{9}\ \frac{N\,m^2}{C^2} \approx 9 \cdot 10^{9}\ \frac{N\,m^2}{C^2}$$

$$k = \frac{1}{4\pi\epsilon_0}$$

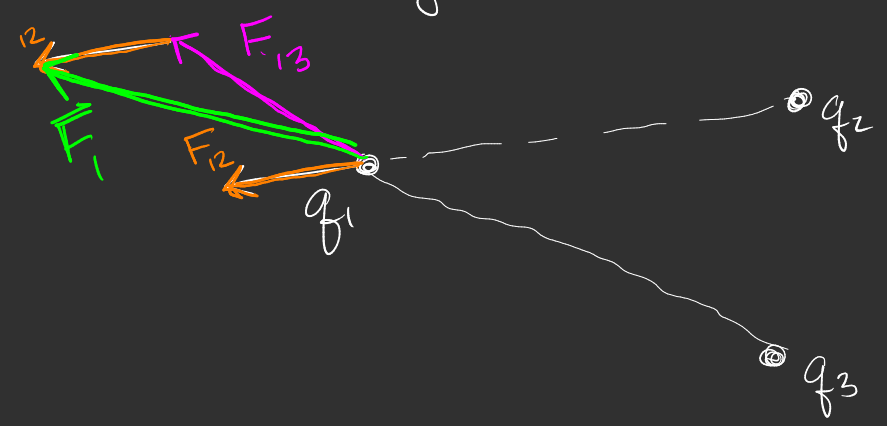
permittivity of free space

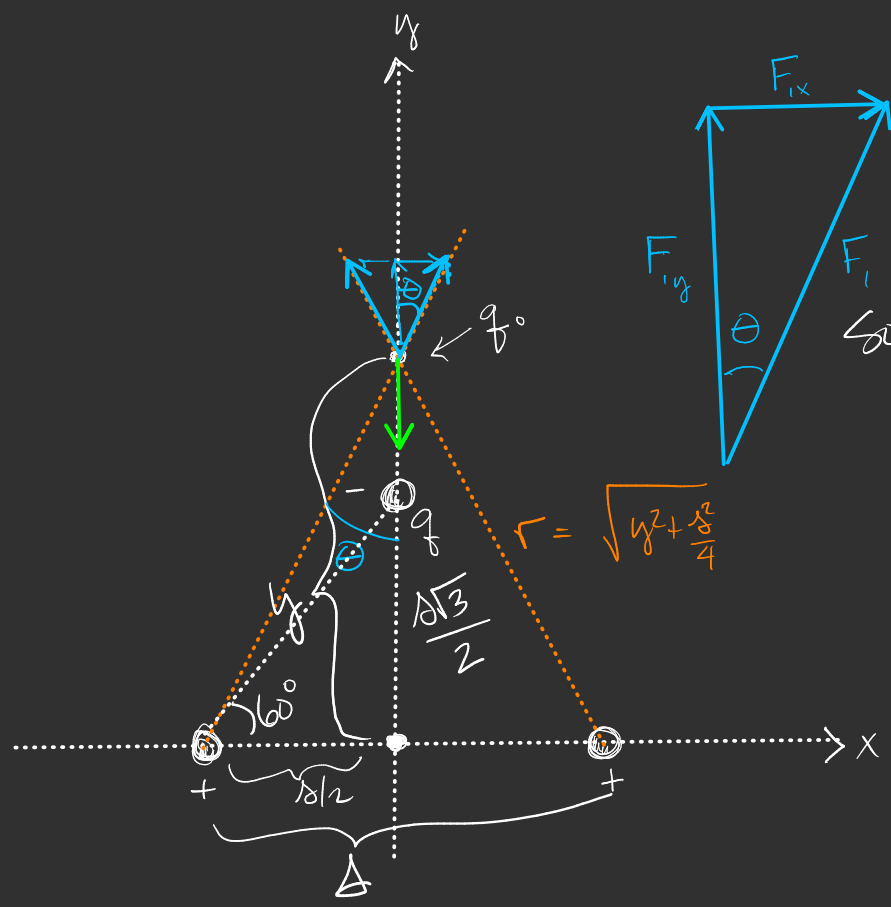
$$\epsilon_0 = 8.85 \cdot 10^{-12}\ \frac{C^2}{N\,m^2}$$

What about more than two charges?

→ Superposition

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13}$$





SOHCAHTOA

$$\cos\theta = \frac{y}{\sqrt{y^2 + \frac{\delta^2}{4}}}$$

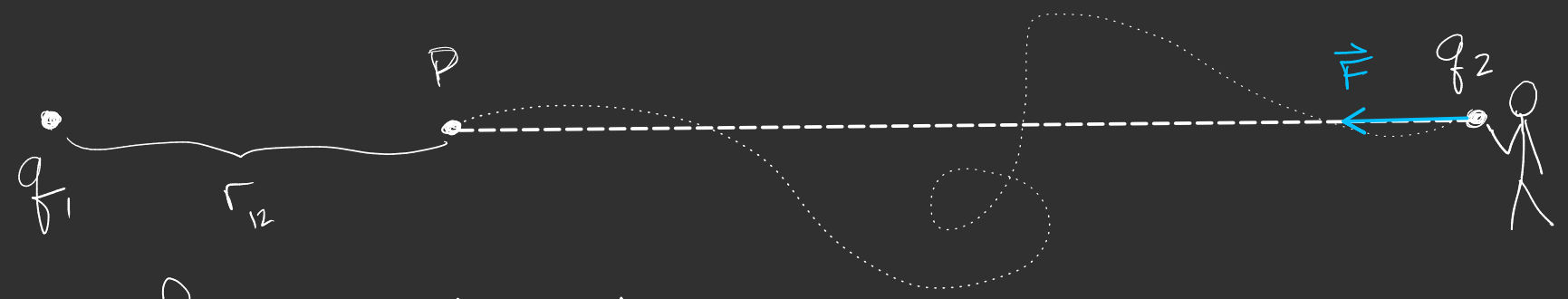
$$0 = F = 2 \cdot \frac{k\cancel{q}q_0}{y^2 + \frac{\delta^2}{4}} \cdot \frac{y}{\sqrt{y^2 + \frac{\delta^2}{4}}} - \frac{k\cancel{q}q_0}{\left(y - \frac{\delta\sqrt{3}}{2}\right)^2}$$

$$\frac{1}{\left(y - \frac{\delta\sqrt{3}}{2}\right)^2} = \frac{2 \cdot y}{\left(y^2 + \frac{\delta^2}{4}\right)^{3/2}}$$

$$y = \frac{\left(y^2 + \frac{\delta^2}{4}\right)^{3/2}}{2\left(y - \frac{\delta\sqrt{3}}{2}\right)^2}$$

$\delta = 1$ $\quad y = 3.1, 0.29, 0.12$?

Work + Energy



$$W = \int \text{force} \cdot \text{displacement}$$

$$W = \int \vec{F} \cdot d\vec{r}$$

Ex. move radially

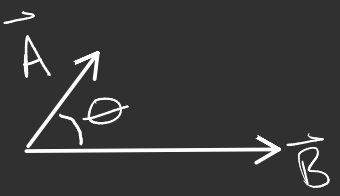
$$W_{ME} = \int_{\infty}^{r_{12}} -\frac{kq_1q_2}{r^2}\,\hat{r} \cdot d\vec{r}$$

$$= -kq_1q_2 \int_{\infty}^{r_{12}} r^{-2}\,dr$$

$$= -kq_1q_2 \left(\frac{(-1)}{r}\right)\Bigg|_{\infty}^{r_{12}}$$

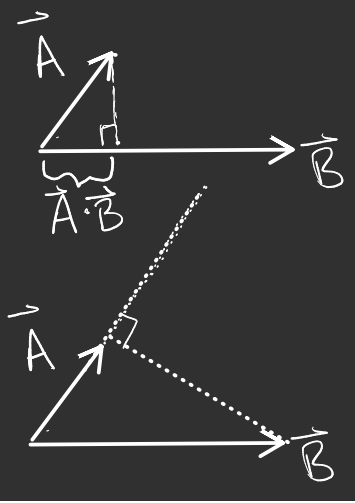
$$\boxed{W_{ME} = \frac{kq_1q_2}{r_{12}}}$$

$$\vec{A} \cdot \vec{B} = |A||B| \cos\theta$$

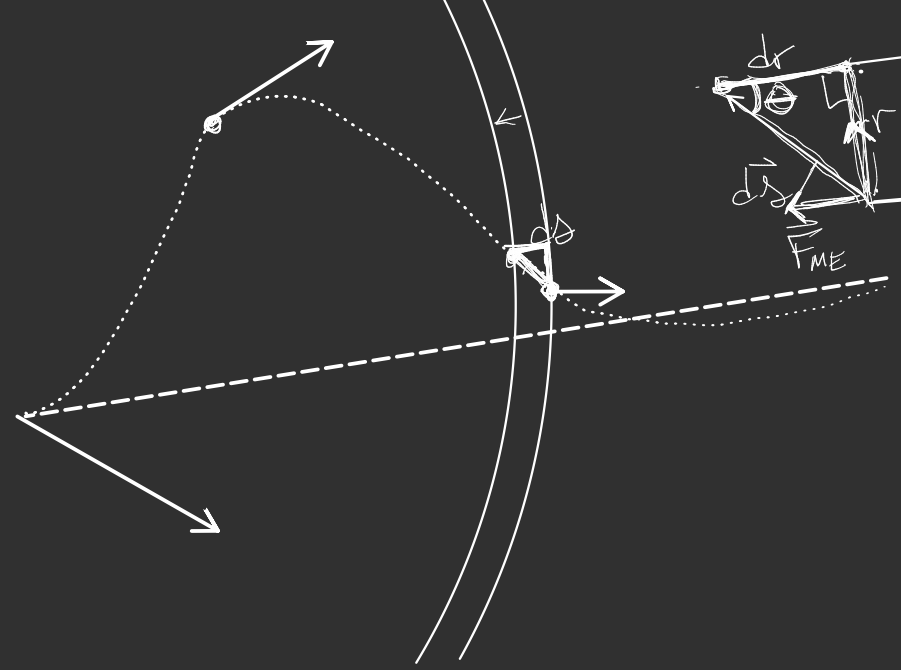


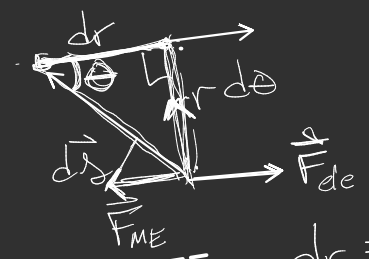
$$\vec{A} = (A_x\hat{x} + A_y\hat{y} + A_z\hat{z})(B_x\hat{x} + B_y\hat{y} \ldots$$

$$\vec{A} \cdot \vec{B} = A_xB_x + A_yB_y + A_zB_z$$



$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$





$$dW = \vec{F} \cdot d\vec{s}$$

$$dr = ds\cos\theta$$

$$\frac{dr}{\cos\theta} = ds$$

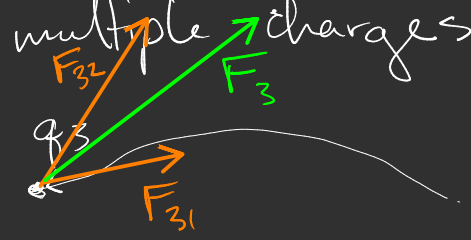
$$\vec{F} \cdot d\vec{s}$$

$$F\,ds\cos\theta$$

$$F\,dr$$

$$\vec{\nabla} \cdot \vec{F} = 0$$

What about multiple charges?



q_1

q_2

$$\vec{F}_3 = \vec{F}_{32} + \vec{F}_{31}$$

$$W_3 = \int \vec{F}_3 \cdot d\vec{r}$$

$$= \int \left(\vec{F}_{32} + \vec{F}_{31}\right) \cdot d\vec{r}$$

$$= \int \vec{F}_{32} \cdot d\vec{r} + \int \vec{F}_{31} \cdot d\vec{r}$$

$$W_3 = W_{32} + W_{31}$$

So what is the total work?

$$W_T = W_{21} + W_{31} + W_{32}$$

$$W_T = k\left(\frac{q_1 q_2}{r_{21}} + \frac{q_1 q_3}{r_{31}} + \frac{q_2 q_3}{r_{32}}\right)$$

Since F_e is conservative, W to build the charges → potential energy

Remington: $W_{ext} = U_{electric}$

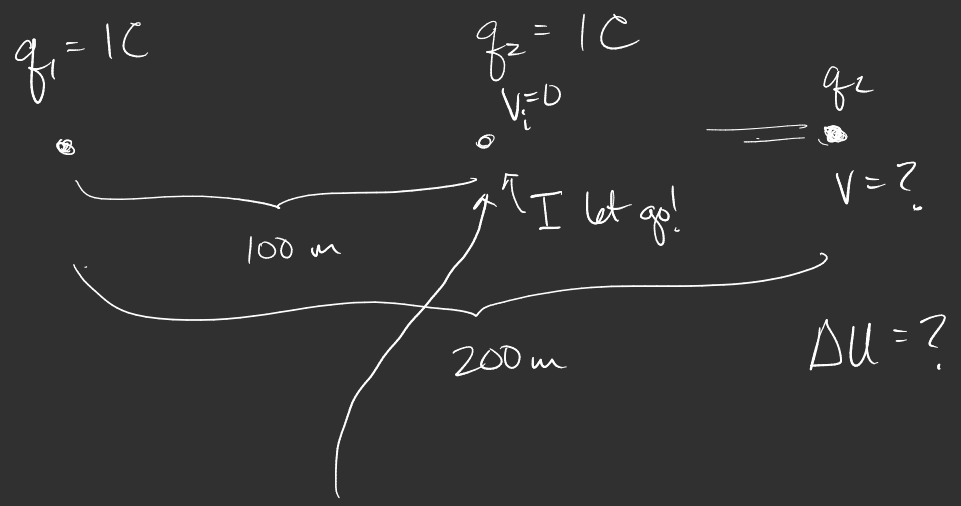
Books: $W_{charges} = -U_{electric}$

conservation of energy →

$$E_i + \underbrace{\Delta E}_{W_{ext}} = \underbrace{E_f}_{K_f + U_f}$$

$$K_i + U_i + \underbrace{W_{ext}}_{\text{non conservative}} = K_f + U_f$$

$$W_{ext} = \Delta K + \Delta U$$



$$W_{ext} = 0$$

$$0 = \Delta K + \Delta U$$

$$\boxed{\Delta K = -\Delta U}$$

$$U(r) = W(\infty \to r)$$

$$U(r) = \frac{k q_1 q_2}{r}$$

$$\Delta U = U(200\,m) - U(100\,m)$$

$$= \frac{9 \cdot 10^9 \cdot 1C \cdot 1C}{200} - \frac{9 \cdot 10^9 \cdot 1C \cdot 1C}{100}$$

$$= 9 \cdot 10^9 \left(\frac{1}{200} - \frac{1}{100} \right)$$

$$= -4.5 \cdot 10^9\ J$$

$$K_i = 0$$

$$\Delta K = 4.5 \cdot 10^9\ J \rightsquigarrow K_f = \frac{1}{2} m v_f^2$$

$$\Delta K = K_f - K_i$$

$$= \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$= \frac{1}{2} m \left(v_f^2 - v_i^2 \right)$$

Electric Field

$$\vec{F} = q_0 \vec{E}$$

$$\vec{E} = \frac{\vec{F}}{q}$$

force per unit charge.

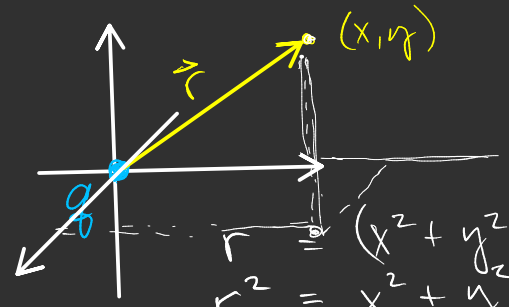
for a point charge

$$\vec{E}(\vec{r}) = \frac{kq}{r^2}\hat{r}$$

state in cartesian coords.

source at origin

position vector $\rightarrow$ $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$



$$r = (x^2 + y^2 + z^2)^{1/2}$$

$$r^2 = x^2 + y^2 + z^2$$

$$\hat{r} = \frac{x\hat{x} + y\hat{y} + z\hat{z}}{(x^2 + y^2 + z^2)^{1/2}}$$

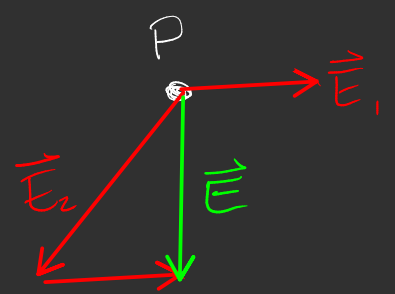
for a charge distribution

$$\vec{E}(x,y,z) = \frac{1}{4\pi\epsilon_0}\sum_{j=1}^{N}\frac{q_j}{r_{0j}^2}\hat{r}_{0j}$$

$$\vec{E}(x,y,z) = \frac{kq\cdot(x\hat{x} + y\hat{y} + z\hat{z})}{(x^2 + y^2 + z^2)(x^2 + y^2 + z^2)^{1/2}}$$

$$= \frac{kq\cdot(x\hat{x} + y\hat{y} + z\hat{z})}{(x^2 + y^2 + z^2)^{3/2}}$$

$q_1 = 1C$



$q_2 = -2C$

Continuous charge distribution

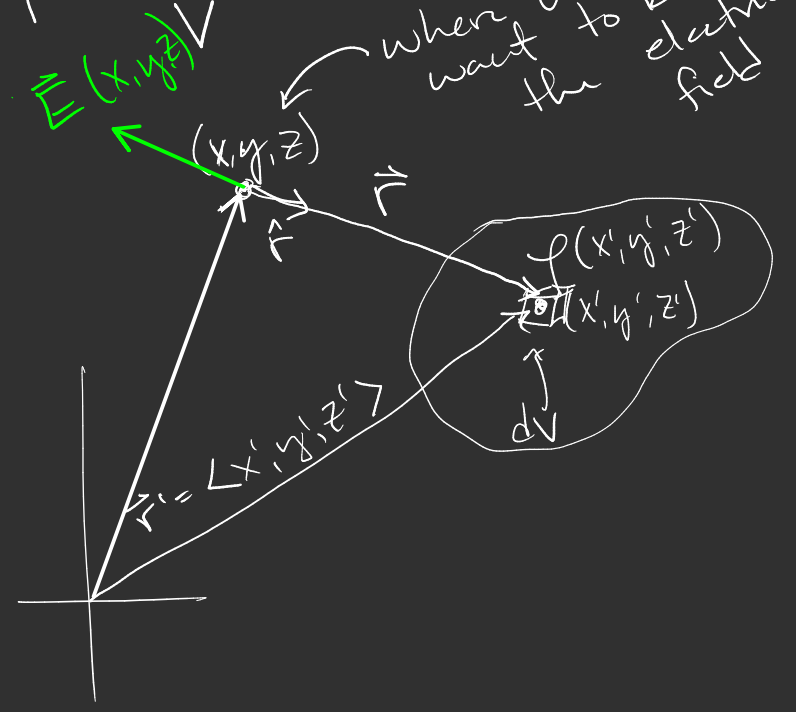
$$\rho \longrightarrow \rho(x', y', z')$$

charge density
charge per volume

$$dq = \rho(x', y', z')\, dV$$

volume element

$$\rho = \frac{Q}{V}$$



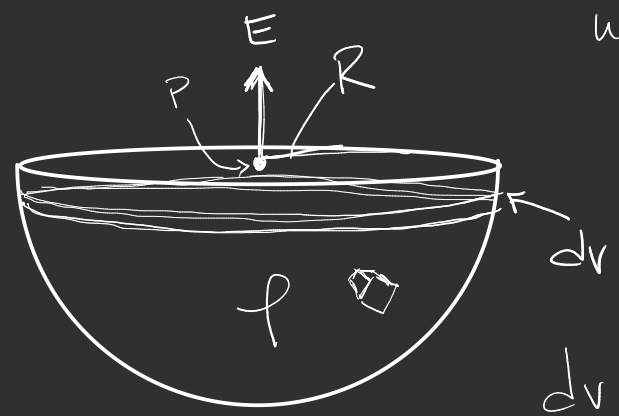
$$\vec{E}(x,y,z) = \frac{1}{4\pi\epsilon_0} \sum_{j=1}^{N} \frac{q_j}{r_{0j}^2} \hat{r}_{0j}$$

$$\Downarrow$$

$$\vec{E}(x,y,z) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(x',y',z')\,\hat{r}\, dV'}{r^2}$$

cartesian:

$$dV = dx\, dy\, dz$$



uniform charge density → ρ

$$E = \frac{1}{4\pi\epsilon_0} \int \frac{\rho \, dV}{r^2}$$

$dv = \pi r^2 dy$ ← polar